DEVELOPER ROLE

SET B — 100 MCQs

Practice assessment · Java, DSA, SQL/DBMS, web, Spring Boot, security, Git/testing, OS and system design.

Instructions: Select one best answer per question. Suggested time: 75 minutes. One mark per question; no negative marking in this practice version.

1. Dynamic programming is especially useful when subproblems are what?

A. Only sorted
B. Limited to graphs
C. Overlapping with reusable results
D. Always independent

2. Which keyword refers to the current Java object?

A. this
B. current
C. self
D. super

3. Binary search on a sorted array takes what worst-case time?

A. O(1)
B. O(log n)
C. O(n)
D. O(n^2)

4. A balanced BST typically supports search in what time?

A. O(n^2)
B. O(log n)
C. O(1) guaranteed
D. O(2■)

5. Which Java type stores key-value pairs?

A. HashMap
B. ArrayList
C. PriorityQueue
D. HashSet

6. What helps prevent SQL injection?

A. Concatenating user input into SQL
B. Parameterized queries
C. Exposing SQL errors
D. Disabling indexes

7. Average lookup in a well-distributed hash table is typically?

A. O(n)
B. O(log n)
C. O(n log n)
D. O(1)

8. Which HTTP status means authentication is missing or invalid?

A. 400
B. 401
C. 500
D. 403

9. Which technique is commonly used to find connected components?

A. Graph traversal
B. Binary search only
C. Hashing strings only
D. Insertion sort

10. Insert at the head of a linked list when head is known takes?

A. $O(n^2)$
B. O(n)
C. O(1)
D. O(log n)

11. Which structure efficiently retrieves the highest-priority item?

A. Stack
B. Basic queue
C. Priority queue
D. Singly linked list

12. Which algorithm finds a minimum spanning tree?

A. KMP
B. Binary search
C. Kruskal's algorithm
D. Floyd cycle detection

13. What does GROUP BY do?

A. Renames a table
B. Groups rows for aggregation
C. Deletes duplicates
D. Sort alphabetically

14. Which SQL command retrieves rows?

A. SELECT
B. GET
C. OPEN
D. FETCH TABLE

15. What does @Transactional help manage?

A. Network routes
B. HTML layout
C. JWT keys only
D. Database transaction boundaries

16. Which layer should normally handle HTTP request mapping?

A. Repository
B. Controller
C. Database index
D. Build tool

17. What is quicksort's worst-case time complexity?

A. O(n log n)
B. O(log n)
C. $O(n^2)$
D. O(n)

18. Which command creates and switches to a branch?

A. git fetch --delete
B. git switch -c branch-name
C. git stash pop
D. git reset --hard

19. Which algorithmic strategy chooses a locally optimal option at each step?

A. Dynamic linking
B. Hashing
C. Greedy
D. Normalization

20. Which join returns only matching rows from both tables?

A. CROSS JOIN
B. LEFT JOIN
C. FULL OUTER JOIN
D. INNER JOIN

21. Which clause filters aggregated groups?

A. HAVING
B. DISTINCT
C. WHERE
D. LIMIT

22. Which Java keyword prevents class inheritance?

A. private
B. volatile
C. final
D. static

23. Which block commonly runs whether or not an exception occurs?

A. throw
B. assert
C. finally
D. catch only

24. How should passwords be stored?

A. Reversible Base64
B. Unsalted MD5
C. Salted slow hash such as BCrypt/Argon2
D. Plain text

25. Which Java type is designed for mutable strings without synchronization?

A. Character
B. StringLiteral
C. String
D. StringBuilder

26. What is the purpose of memoization?

A. Sort linked lists
B. Cache results of repeated function calls
C. Increase recursion depth
D. Encrypt variables

27. What is a constructor used for?

A. Handling exceptions only
B. Initializing a new object
C. Starting a thread automatically
D. Destroying an object

28. Which constraint prevents NULL values in a column?

A. DEFAULT
B. FOREIGN KEY
C. CHECK
D. NOT NULL

29. What is the purpose of a database index?

A. Replace primary keys
B. Speed up selected reads at storage/write trade-off
C. Encrypt every row
D. Prevent all duplicates automatically

30. Which technique explores a graph level by level?

A. DFS
B. Backtracking only
C. BFS
D. Heapify

31. Which file commonly holds Spring Boot configuration?

A. application.properties
B. index.css
C. routes.sql
D. pom.lock

32. Normalization primarily reduces what?

A. Redundancy and update anomalies
B. Query syntax
C. Indexes
D. All joins

33. Which media type represents JSON?

A. text/sql
B. image/java
C. application/json
D. application/java

34. What is true about Java interfaces?

A. Interfaces require constructors
B. Interfaces cannot have default methods
C. A class can implement multiple interfaces
D. A class can extend multiple classes

35. What does COUNT(*) count?

A. Only non-null values in one column
B. Only distinct rows
C. Rows in the result set
D. Only primary keys

36. Which Spring annotation marks a REST controller?

A. @AutowiredOnly
B. @RestController
C. @EntityManager
D. @ServletTable

37. What does a foreign key enforce?

A. Compression
B. Sorting
C. Encryption
D. Referential integrity

38. What does a generic type parameter provide?

A. Runtime encryption
B. Automatic persistence
C. Thread scheduling
D. Compile-time type safety

39. Which command removes all rows while retaining table structure?

A. ALTER
B. RENAME
C. TRUNCATE
D. DROP

40. What is the default value of an instance int field in Java?

A. Uninitialized random value
B. -1
C. 0
D. null

41. What is dependency injection?

A. Supplying an object's dependencies externally
B. Converting Java to SQL
C. Creating every dependency inside a class
D. Encrypting method names

42. Which structure supports the call stack used by recursion?

A. Disjoint set
B. Stack
C. Queue
D. Hash table

43. Which HTTP method is commonly used for partial updates?

A. OPTIONS
B. PATCH
C. GET
D. TRACE

44. Which is a checked Java exception?

A. ArithmeticException
B. ArrayIndexOutOfBoundsException
C. NullPointerException
D. FileNotFoundException

45. What is the Java application entry-point method?

A. init()
B. start()
C. run()
D. public static void main(String[] args)

46. What does a JWT signature provide?

A. Confidentiality of all claims
B. Password storage
C. Infinite token lifetime
D. Integrity and authenticity verification

47. Which clause filters rows before grouping?

A. ORDER BY
B. HAVING
C. GROUP BY
D. WHERE

48. Which collection preserves insertion order and allows duplicates?

A. HashMap
B. HashSet
C. TreeSet
D. ArrayList

49. Which data structure follows LIFO?

A. Stack
B. Queue
C. Heap
D. Graph

50. Which data structure is suitable for union/find connectivity queries?

A. Disjoint-set union
B. Trie
C. Stack
D. Deque only

51. Which Spring annotation maps GET requests?

A. @PutMapping
B. @GetMapping
C. @PostMapping
D. @DeleteMapping

52. What does HTTPS add to HTTP?

A. Automatic database caching
B. Guaranteed zero latency
C. Mandatory compression
D. TLS transport security

53. What does JVM stand for?

A. Java Variable Manager
B. Java Verified Module
C. Java Virtual Machine
D. Joint Version Model

54. Which key uniquely identifies a table row?

A. Primary key
B. Foreign key
C. Secondary index
D. Duplicate key

55. What does COMMIT do?

A. Persist transaction changes
B. Undo a transaction
C. Create an index
D. Delete a database

56. Which interface is the root of the Java collection hierarchy for groups of elements?

A. Comparable
B. Collection
C. Serializable
D. Runnable

57. What is Git primarily used for?

A. Web design
B. Version control and collaboration
C. Running production servers
D. Relational storage

58. Which sorting algorithm is typically stable?

A. Selection sort
B. Heap sort
C. Merge sort
D. In-place quicksort

59. REST is best described as what?

A. CSS framework
B. Architectural style for networked applications
C. Programming language
D. Database engine

60. What does least privilege mean?

A. Share one account
B. Grant only necessary permissions
C. Everyone gets admin access
D. Disable authentication

61. What is SQL injection?

A. Backup method
B. Query optimization
C. Attack manipulating SQL via untrusted input
D. Normalization form

62. What is a database transaction?

A. Logical unit of work committed or rolled back
B. File format
C. User account
D. Index type

63. Which Java collection rejects duplicate elements?

A. Vector
B. HashSet
C. ArrayList
D. LinkedList

64. Which ACID property means all-or-nothing?

A. Durability
B. Atomicity
C. Isolation
D. Consistency

65. Which application layer usually contains business rules?

A. Physical network
B. DNS
C. Service/business
D. Presentation

66. Which HTTP status commonly means resource created?

A. 404
B. 200
C. 301
D. 201

67. Which command changes a table definition?

A. ALTER TABLE
B. UPDATE
C. SELECT
D. COMMIT

68. What does git pull typically do?

A. Rewrites all commit messages
B. Fetches and integrates remote changes
C. Deletes local branches
D. Creates a hosted repository

69. Which index structure is common for database range queries?

A. Call stack
B. Stack
C. Binary heap
D. B-tree/B+ tree

70. What is the time complexity of merging two sorted arrays of total length n?

A. $O(n^2)$
B. O(log n)
C. O(1)
D. O(n)

71. When properly overridden, equals() usually checks what?

A. Class loading order
B. Reference identity only
C. Logical equality
D. Object size

72. What is the typical average time complexity of hash-table insertion?

A. O(1)
B. $O(n^2)$
C. O(log n)
D. O(n log n)

73. Which status code represents an internal server error?

A. 404
B. 302
C. 500
D. 201

74. What is Flyway commonly used for?

A. UI icons
B. Bytecode compression
C. Load balancing
D. Versioning and applying database schema changes

75. Which SQL statement changes existing rows?

A. GRANT
B. ALTER
C. UPDATE
D. CREATE

76. Which protocol provides reliable ordered byte-stream delivery?

A. TCP
B. UDP
C. ARP
D. ICMP

77. Which data structure follows FIFO?

A. Queue
B. Tree
C. Set
D. Stack

78. Which standard Java class is immutable?

A. String
B. ArrayList
C. StringBuilder
D. StringBuffer

79. Which SQL operator checks whether a value matches any value in a list?

A. LIKE ONLY
B. IN
C. IS NULL
D. BETWEEN

80. Which keyword explicitly throws an exception?

A. throw
B. catch
C. final
D. throws

81. What is a DTO used for?

A. Start threads
B. Replace every service
C. Exact database engine representation
D. Transfer data across application boundaries

82. What does static mean for a class field?

A. One copy per object
B. The field cannot be read
C. The field is always immutable
D. A class-level member shared by instances

83. What does CORS govern?

A. Cross-origin browser requests
B. Password hashing
C. CPU scheduling
D. Database migrations

84. Why paginate API results?

A. Replace authorization
B. Prevent all errors
C. Return large result sets in manageable chunks
D. Encrypt payloads

85. How can a linked-list cycle be detected with O(1) extra space?

A. Counting sort
B. Binary search
C. Floyd slow/fast pointers
D. Topological sort

86. What is the purpose of try-with-resources?

A. Automatically close AutoCloseable resources
B. Retry failed requests
C. Create threads
D. Suppress all exceptions

87. Which keyword makes a variable's visibility changes observable across threads but does not make compound operations atomic?

A. transient
B. strictfp
C. volatile
D. native

88. What is method overriding?

A. Calling a static method
B. Subclass implementation of an inherited method
C. Declaring multiple constructors
D. Changing a variable type

89. Which access modifier gives the broadest visibility?

A. public
B. private
C. protected
D. package-private

90. Which HTTP method is generally used to retrieve a resource?

A. PATCH
B. GET
C. DELETE
D. POST

91. Which OOP principle hides internal state behind methods?

A. Abstraction
B. Encapsulation
C. Polymorphism
D. Inheritance

92. Where should production secrets be kept?

A. Frontend bundle
B. Secret manager/protected environment
C. Committed source code
D. Public README

93. Which algorithm finds shortest paths in an unweighted graph?

A. Bubble sort
B. Binary search
C. BFS
D. DFS

94. Which BST traversal yields sorted keys?

A. Preorder
B. Postorder
C. Reverse level order
D. Inorder

95. What is method overloading?

A. Subclass replacement of a method
B. Hiding a field
C. Changing a return value only
D. Same name with different parameter lists

96. Which map implementation generally does not guarantee iteration order?

A. EnumMap
B. TreeMap
C. LinkedHashMap
D. HashMap

97. An adjacency-list graph representation typically uses what space?

A. $O(V^2)$ always
B. O(log V)
C. $O(E^2)$
D. O(V + E)

98. What does DNS do?

A. Create database keys
B. Encrypt HTTP traffic
C. Compile Java
D. Resolve domain names to IP addresses

99. Which structure is commonly used for efficient prefix searches?

A. Binary heap
B. Trie
C. Stack
D. Queue

100. What does a repository usually abstract?

A. Data access operations
B. DNS configuration
C. Browser history
D. CSS rendering